



A HYBRID AI PIPELINE FOR THE CLASSIFICATION & SEGMENTATION OF MARINE OIL SPILLS

Detecting oil spills in Sentinel-1 SAR radar with deep learning

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01

PROBLEM & SOLUTION



THE PROBLEM & OUR SOLUTION



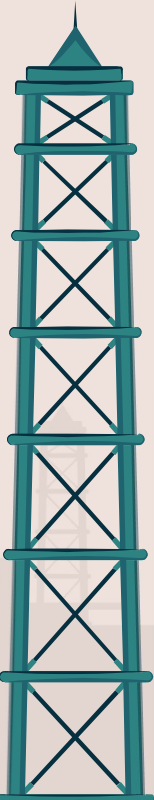
THE PROBLEM

The problem is not only detecting dark regions in SAR images. The real challenge is separating true oil spills from look-alikes such as low wind, rain, and natural films. On top of that, oil covers only about 1% of the pixels, so normal accuracy can be misleading. This is why I treat the task as semantic segmentation and evaluate using oil-class IoU.



OUR SOLUTION

Our solution is to use Sentinel-1 SAR radar images and treat oil-spill detection as semantic segmentation, so the system predicts exactly which pixels belong to oil. Instead of relying on one model, I compare a family of U-Net-based models under the same fair setup, then deliver the best compact model as a reproducible prototype.



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LITERATURE & GAPS



LITERATURE REVIEW

CLASSIC BASELINES

U-Net, LinkNet, PSPNet, DeepLabv3+. The Krestenitis baseline reaches about 0.53 oil-class IoU.

ATTENTION U-NETS

Add attention to focus on oil and ignore clutter (e.g. GSCAT-UNET).

SHAPE-AWARE NETS

Use the long, thin shape of slicks (e.g. SAGPNet, strip attention).

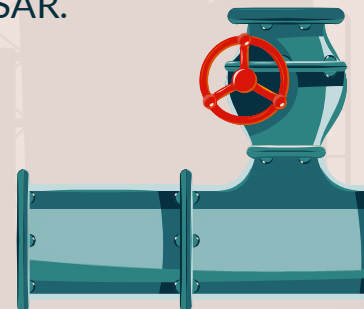
TRANSFORMER / HYBRID

Add global reasoning over the whole image (e.g. STDPNet, CNN-Transformer).

OPTICAL / UAV PORTS

GSSNet, SGDBNet report over 0.9 — but on easy scenes, not hard SAR.

Model name	Accuracy	Sensitivity	Precision	F1-score	IoU
DL-ViT-1	95.64	68.11	81.24	74.10	58.86
DL-ViT-2	96.45	72.09	86.12	78.48	64.59
DL-ViT-3	95.40	70.21	77.43	77.19	58.28
FC-ViT-1	95.67	70.00	80.21	74.76	59.69
FC-ViT-2	95.55	68.81	79.81	73.90	58.61
U-ViT	94.79	66.18	74.15	69.94	53.78
DeepLabV3+	95.71	69.50	81.64	75.08	60.10
FC-DenseNet	95.56	68.75	79.99	73.94	58.66
U-Net	94.17	49.48	78.99	60.85	43.73



GAPS IN PRIOR WORK

TANGLED IMPROVEMENTS

Prior work changes too many things at once, so we cannot know what actually caused the improvement.

EFFICIENCY TRADE-OFF

The strongest results often rely on large pretrained backbones, which are slow and memory-hungry to train and run.

UNTESTED GATES

Many pipelines add a classifier before segmentation, but do not test whether it improves the full pipeline.



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RESEARCH QUESTIONS



RESEARCH QUESTIONS

RQ1 · DOES THE GATE HELP?

Does adding an image-level oil / no-oil classifier in front of the segmenter improve the whole pipeline, or hurt it?

RQ2 · CAPABILITY VS CAPACITY

With all training settings kept equal, which architectural ideas truly improve oil-spill segmentation — and does capability matter more than model size?

RQ3 · ENSEMBLE & TTA

Can averaging the two best models and adding test-time augmentation push accuracy higher — and is the extra cost worth it?



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METHODOLOGY & SYSTEM



PHASE 1 — THE CLASSIFIER GATE



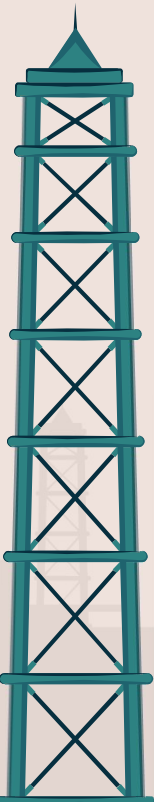
WHAT IT IS

A lightweight image-level oil / no-oil classifier (~5M params): a residual CNN with CBAM attention and an ASPP block, on the 3-channel input (VV, VH, VV-VH). It gives one oil-or-not decision per tile.



WHY A GATE

Most of a scene is open water, so a cheap oil / no-oil check could skip clean tiles before the heavier segmenter — saving compute and cutting false alarms. It is an **OPTIONAL** first stage; whether it actually helps is tested in RQ1.



EACH MODEL — ENCODER · BOTTLENECK · DECODER

Model	Encoder	Bottleneck	Decoder
M1 · Plain U-Net	Double-conv + max-pool	Double-conv	Double-conv · plain skip-concat
M2 · Residual-Attention	Residual blocks	Residual block	Residual · attention-gated skips
M3 · Shape-Transformer	Residual + multi-scale shape block	Transformer + strip attention	Residual · attention-gated skips
M4 · ASPP+CBAM	Residual + multi-scale shape block	Transformer + strip + ASPP + CBAM	Residual · attention-gated skips
M5 · DMoE ★	Residual + dual attention	Mixture-of-experts (4 experts)	Residual · attention-gated skips + edge head

Input: 2 channels (VV, VH) for M1–M3; 3 channels (VV, VH, VV–VH) for the polarimetric M4 & M5. The bottleneck is where each model's signature mechanism lives.

METHODOLOGY

ONE RECIPE

Same dataset, sampler, augmentation, AdamW optimiser and plain BCE for all five models. M5 adds only a small auxiliary edge term.

ONLY ARCHITECTURE CHANGES

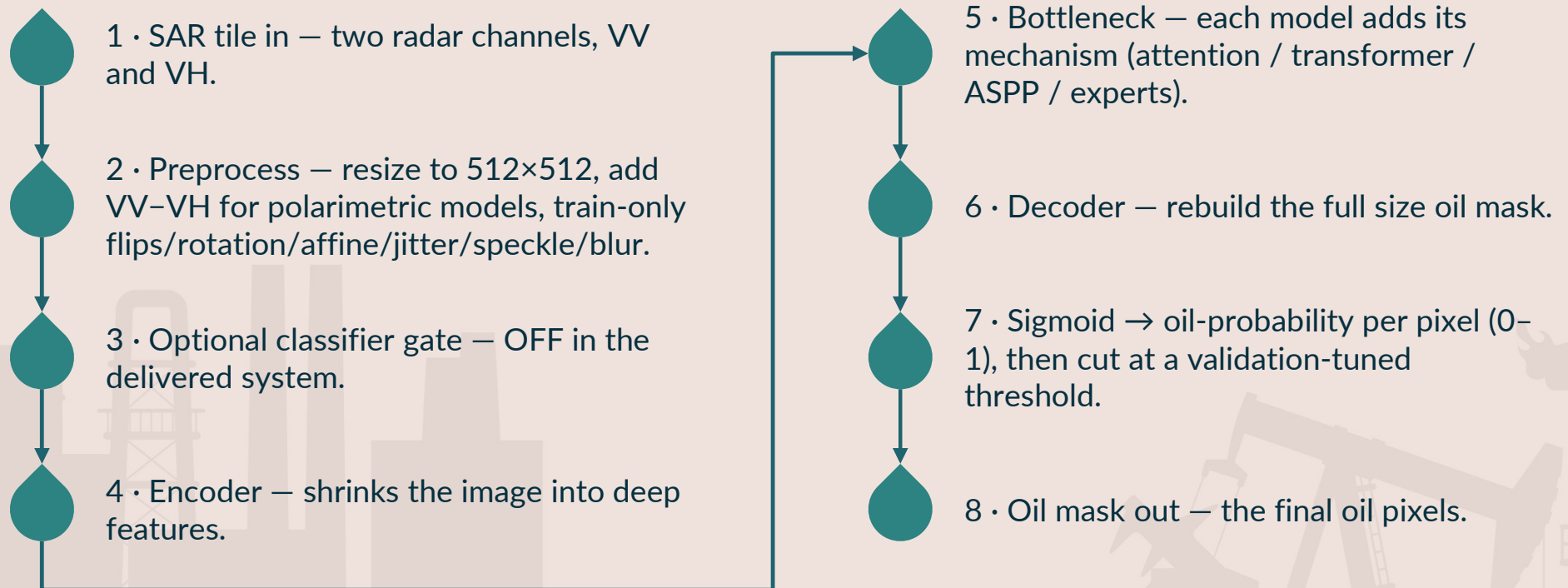
So any difference in score comes from the design — not from loss tricks or extra model size.

HONEST TEST

Threshold picked on validation; tested once on the held-out Part III; 95% bootstrap confidence intervals.



SYSTEM ARCHITECTURE — INPUT TO OUTPUT



DATASET & SPLIT

450

held-out test tiles (Part III)

2,056

training tiles (Parts I + II)

514

validation tiles



PREPROCESSING & CLASS IMBALANCE



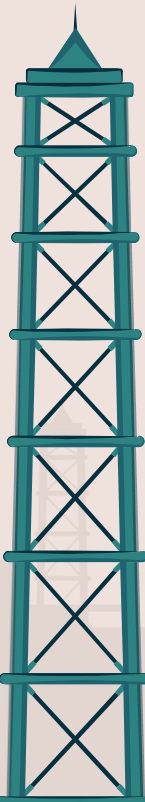
PREPROCESSING

512×512 tiles; VV & VH in decibels. No extra speckle filtering (the data is already filtered). Channel-wise standardisation; polarimetric models add a VV-VH channel. Train-only augmentation: flips, rotation, affine, jitter, speckle, blur.



THE 1% PROBLEM

Oil is only ~1% of pixels, so a lazy model could call everything 'water'. Fix: a weighted sampler feeds tiles at 55% oil / 30% look-alike / 15% clean — keeping the loss plain and identical for every model.



ENGINEERING CHALLENGES — HANDLED



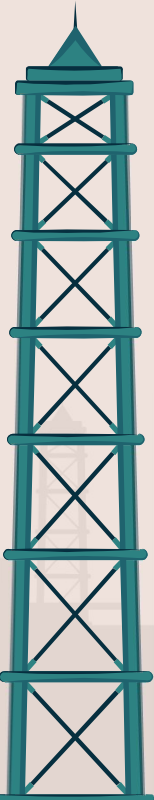
FITTING IT IN MEMORY

The dataset is large (~127 GB), so we hosted it on Google Drive and loaded tiles into Colab (~57 GB RAM, 23.7 GB GPU). Caching tiles in RAM kept filling memory and restarting the session, so we cached in float16 and used gradient checkpointing + batch 4 for the largest models.



SURVIVING RESTARTS

A 100-epoch run outlasts a cloud session that can time out or crash. We save a full checkpoint (model + optimiser + scheduler) every epoch and auto-resume from it — so an interrupted session continues from the exact epoch, with no training lost.

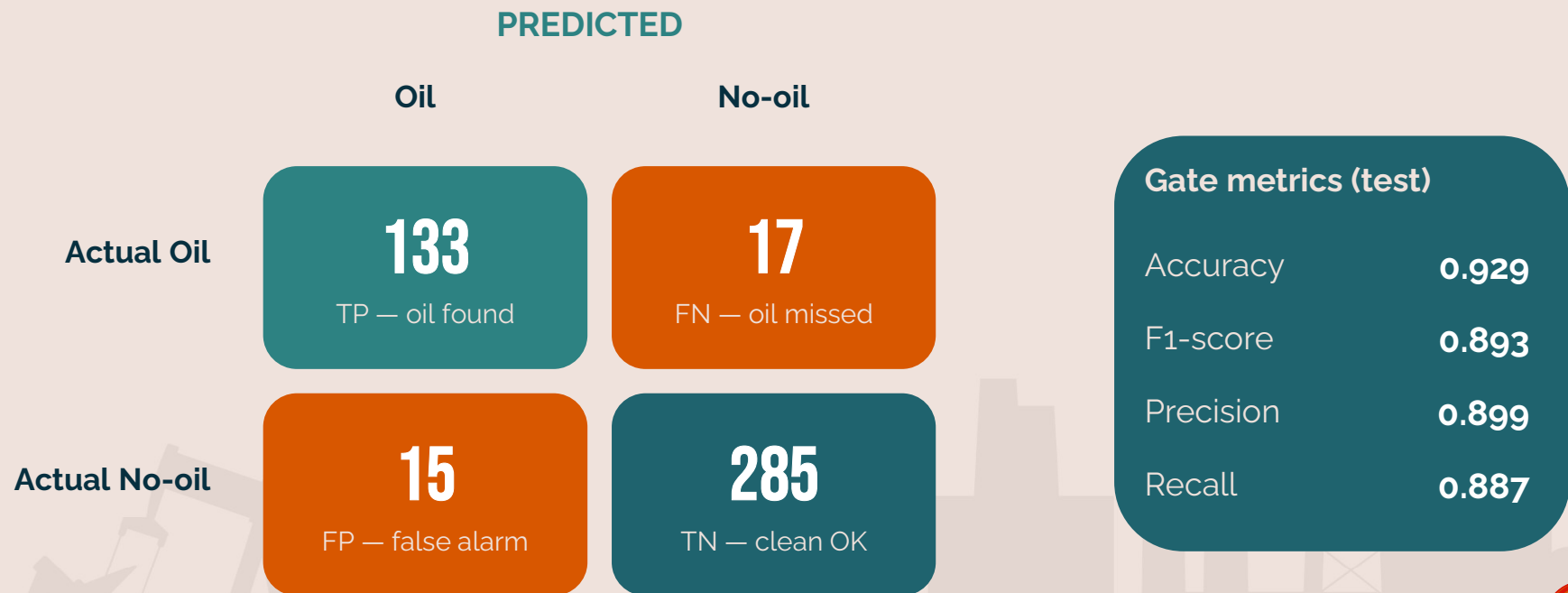


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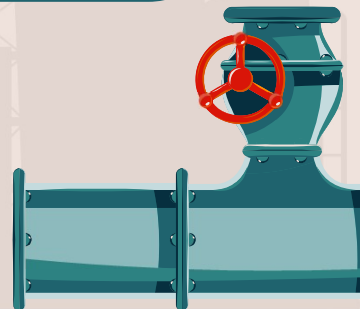
RESULTS & ANALYSIS



RQ1 — DOES THE CLASSIFIER GATE HELP?



Great on its own — but as a hard gate it drops the 17 real oil tiles, lowering recall, so it is kept OPTIONAL and OFF in the delivered system.

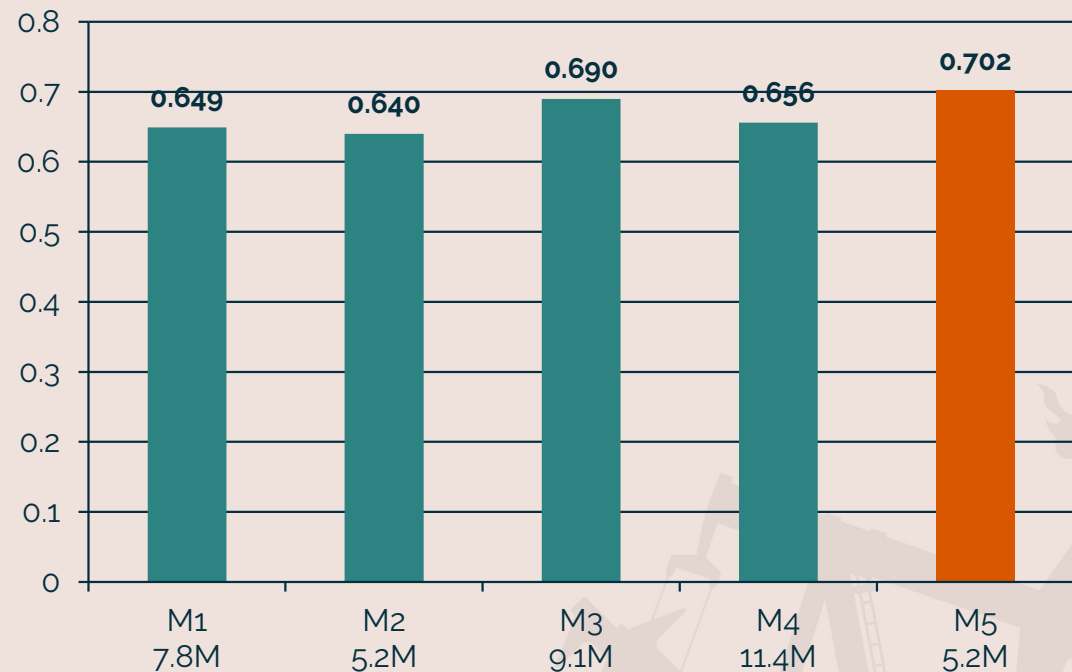


RQ2 — CAPABILITY BEATS CAPACITY

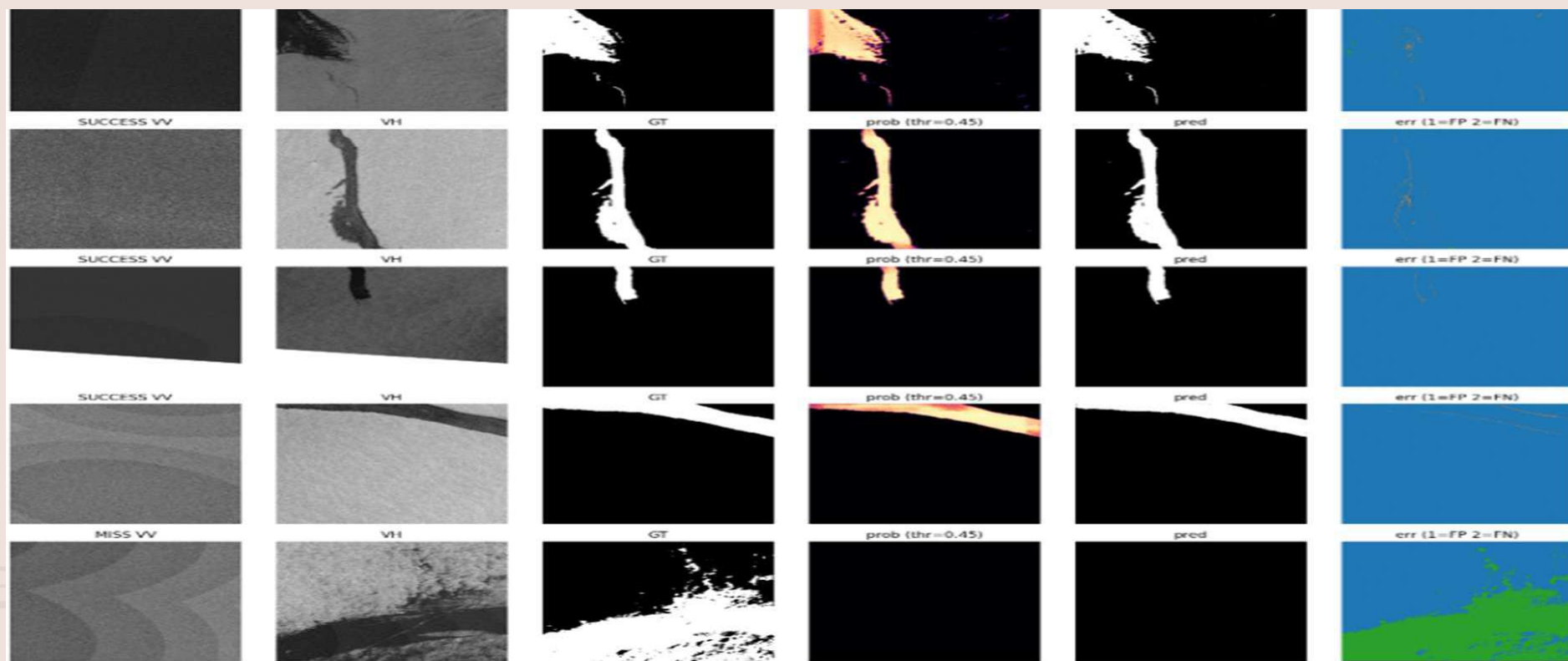
WHAT IT SHOWS

Oil-class IoU on the held-out test set — each model's size (params) is shown under its bar. The DMoE (M5, 5.2M) is the best, beating the 11.4M M4. Bigger is not better; capability wins.

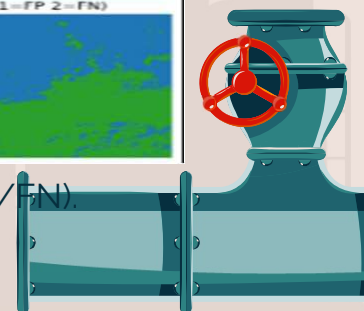
M1 Plain • M2 Res-Att • M3 Shape-Trans
• M4 ASPP+CBAM • M5 DMoE



SEGMENTATION QUALITY – QUALITATIVE EXAMPLES



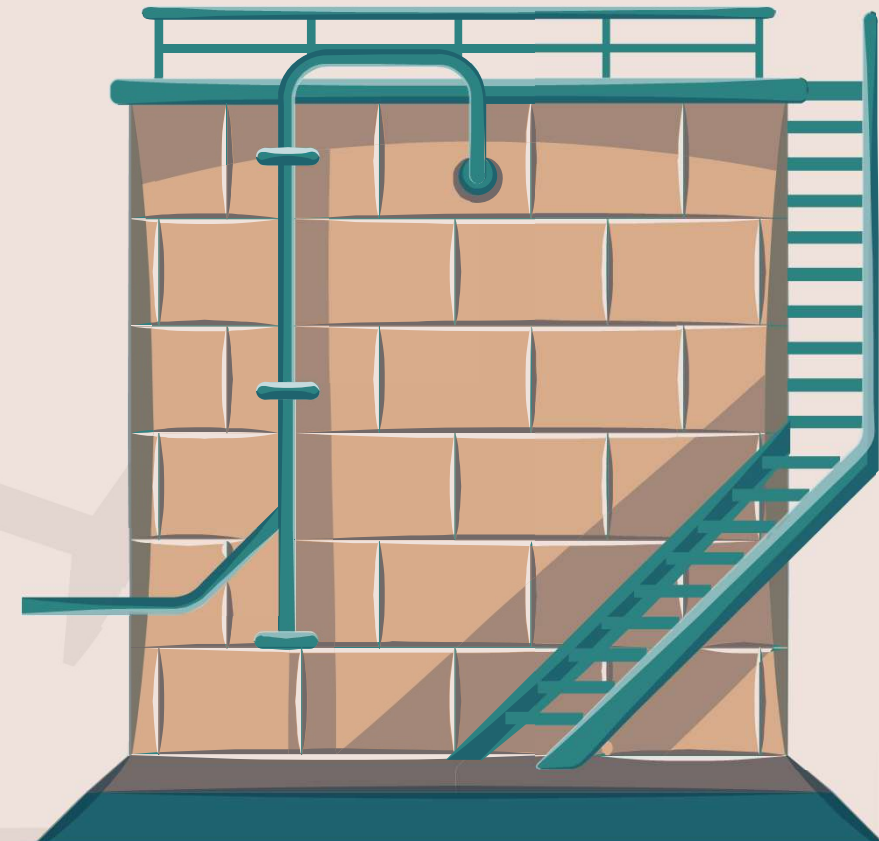
Each row: SAR input (VV, VH) · ground-truth mask · predicted probability · predicted oil mask · error (FP/FN).



0.7021

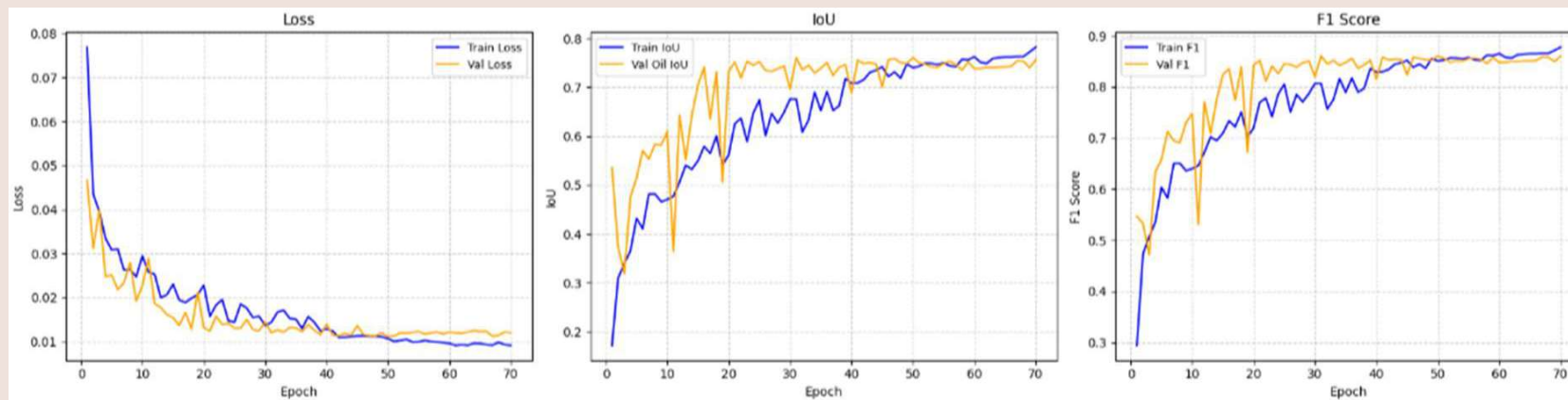
DMoE oil-class IoU on the test set — the best model, and one of the smallest (5.2M).

Model	Architecture	Params (M)	IoU	95% CI
Model 1	Plain U-Net	7.8	0.649	[0.606, 0.687]
Model 2	Residual-Attention	5.2	0.640	[0.598, 0.677]
Model 3	Shape-Transformer	9.1	0.690	[0.648, 0.732]
Model 4	+ASPP+CBAM (pol.)	11.4	0.656	[0.611, 0.699]
Model 5	DMoE dual-attention	5.2	0.702	[0.656, 0.748]

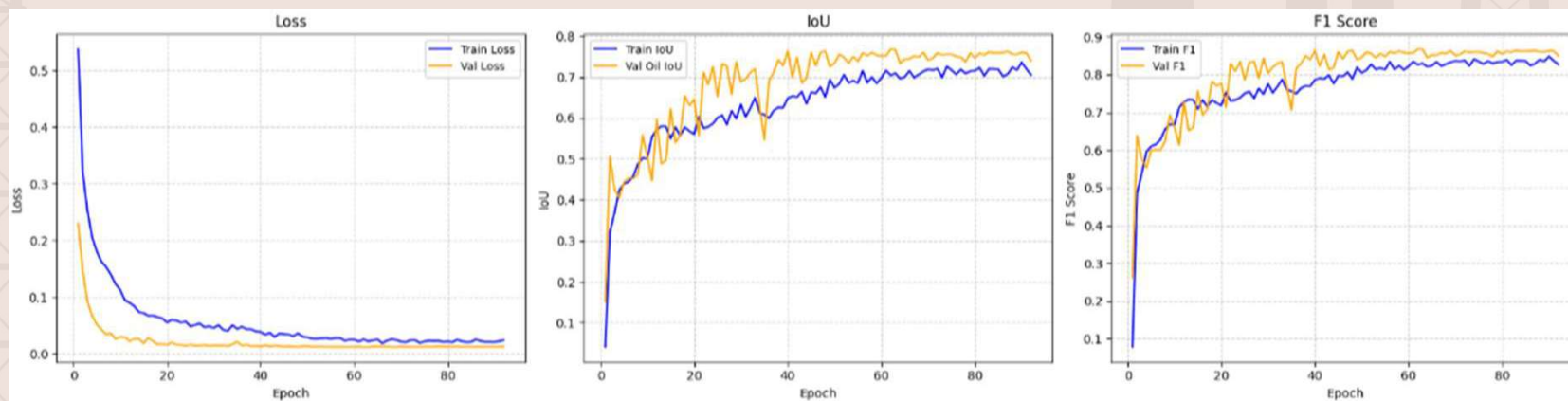


LEARNING CURVES — MODEL 3 & MODEL 5

Model 3 · Shape-Transformer



Model 5 · DMoE (delivered)



RQ3 — ENSEMBLE & TEST-TIME AUGMENTATION

0.72

ensemble + 4-way TTA (best)

0.70

single DMoE

0.71

ensemble (M3 + M5)



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CONCLUSION & FUTURE



CONCLUSION & CONTRIBUTIONS

BEST MODEL

DMoE: 0.7021 IoU.

CAPABILITY > CAPACITY

A 5.2M model beats an 11.4M one — design wins, not size.

FAIR COMPARISON

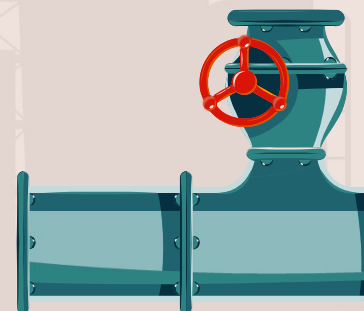
All five models share one recipe, so differences come from architecture alone.

GATE TESTED, NOT ASSUMED

Showed a hard classifier gate hurts the pipeline.

REPRODUCIBLE

Fixed seed, fixed split, saved checkpoints. A research prototype.



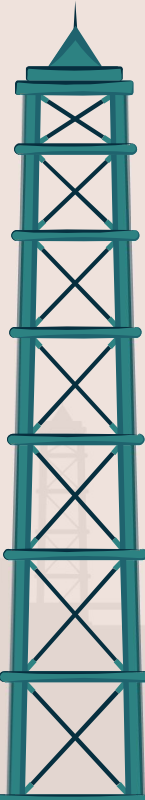
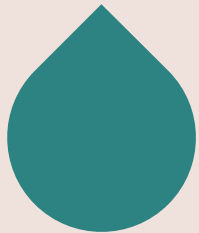
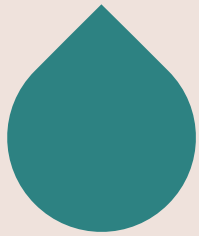
LIMITATIONS & FUTURE WORK

LIMITATIONS

Tiles were resized from 2048×2048 to 512×512 , which may remove fine oil-boundary details; The classifier gate missed 17 real oil tiles, so it is unsafe to use it for now.

FUTURE WORK

Merge Model 3 (Shape-Transformer) and Model 5 (DMoE) into one network; train on the full 2048×2048 tiles instead of the 512 resized, to keep fine detail; improve the classifier gate so it never drops a real oil tile; Validate on external SAR datasets.



THANK YOU

Questions & discussion

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